Research Documentation Rules

Working system for efficient papers, talks, and thesis chapters | November 2026

Core principle

Write from the start of every project. Writing exposes what is missing; a folder of plots does not. Capture happens once, in one place, and feeds every output: papers, talks, teaching notes, and the thesis.

The capture layers (in the repo)

- **Daily log.** Dated markdown: what was tried, what broke, what is next. Never polished. Exists to recover context.
- **Results log.** Any result you would put on a slide gets an entry the day it is produced: a figure generated by a committed script, a dated descriptive filename, and three to five sentences (what was run, what it shows, what it means, what is uncertain). No screenshots, no orphan notebook cells.
- **Code documentation.** Docstrings and short design notes written while the reasoning is fresh, especially for implementation choices such as kernel design patterns.

Arc 1: Research question to paper

- Open the Overleaf paper on day one of the project.
- Give it the paper's shape immediately: background, methods, results, discussion headings, a one-line claim under each, and placeholder figures for results you expect to need.
- Messy prose inside that structure is fine. Chronological notes are not; they stay in the daily log.
- Promote a result from the results log into the draft only when it supports a claim. Target: results log within a day, draft within a few weeks.
- When a direction fails, rewrite the claim and cut the dead section into an **attic** file in the same Overleaf project. Keep the draft as the current best version of the argument; the attic preserves material for the thesis.
- Group meeting and conference talks are built from the same outline as the draft. Each improves the other.

Arc 2: Exploration, debugging, and learning

- Side experiments, bug hunts, and learning a new tool or framework get their own clean, dated notebook or markdown note in the repo.
- Write these as if explaining to a labmate: the problem, what you tried, what worked, why. This is the raw material for a teaching segment or an exploratory talk.
- If an exploration produces a result relevant to a paper, it enters the results log and follows Arc 1. Nothing is documented twice.
- Implementation knowledge decays fastest. Document it early; it can become a software or methods paper and a thesis chapter.

Promotion rules (summary)

- Daily log is for you tomorrow. Results log is for you in three months. The draft is for everyone else.
- Slide-worthy result: results log same day.
- Result that supports a claim: paper draft within weeks.
- Reusable know-how: clean note now; talk or tutorial when asked.
- Failed direction: attic, not deleted.

Goal

Four papers before graduating, strong talks along the way, and real research skill built in the process. The system exists to minimize rework, not to maximize notes.

Arc 3: Background understanding

Why it matters. Papers reward results, but every output also needs a precise account of why the work matters and how it works. That account is written once and reused in paper introductions and background sections, the opening slides of every talk, thesis chapters 1 and 2 (often 30 to 50 pages in a physics thesis), and research interviews, where explaining why a method works and why the obvious alternative fails is a standard question. Writing it is also how understanding becomes articulable: the act of explaining a mechanism exposes the parts you only thought you understood. The same skill carries to industry, where good research teams keep internal notes on methods and model assumptions so colleagues can check them.

- **Write only when a trigger fires:** you need to explain it soon (talk, paper section, meeting); you have read the same thing twice because it did not stick; or a question keeps coming back.
- **One note per question, titled as the question.** For example: why does position scatter bias redundant calibration? How does gain error couple foregrounds into the 21 cm power spectrum?
- **One to three pages, in your own words, with citations.** No comprehensive field reviews.
- **Home: the thesis Overleaf project, started now.** Notes go into the background chapters as standalone sections. Rough and disconnected for a year is fine.
- **Seed it with the prelim report.** Split it into question-titled sections and drop them in.
- **Budget:** about one note every week or two. If it crowds out the paper, keep only the first two triggers.